Why is i^i a real number and what does it equal?

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1 Introduction

There are a few different ways to approach this problem. In Section 2 we use the fact that complex conjugation is an automorphism of $\mathbb{C}$. In Section 3 we use the power series expansion of i^i to show that $i^i \in \mathbb{R}$, and in Section 4 we rely on Euler's formula to show that $i^i \in \mathbb{R}$. Section 5 does a bit of arithmetic to show what i^i equals numerically. Finally Section 6 offers a few conclusions.

Before launching into all of this, note that we will make heavy use of the exponential function $\exp(z)$ (and therefore as you might expect, this means heavy use of $\log(z)$). The important point for this discussion is that $\exp(z)$ is injective over $\mathbb{R}$ but not over $\mathbb{C}$. Therefore, if we want to define the logarithm as it's inverse (as we will), we will need to select a subset of $\mathbb{C}$ where the exponential function is injective. In this sense, it is not entirely correct to say that log is a multivalued function; rather, what we have is different logarithm functions depending on the choice of a set where $\exp(z)$, for some $z \in \mathbb{C}$, is injective. Generally this subset is called a principal branch (a principal branch is a function which selects one branch ("slice") of a multi-valued function [4]).

2 Automorphism of $\mathbb{C}$

In this section we look at solving this puzzle using the fact that complex conjugation is an automorphism of $\mathbb{C}$ [2]. For this approach we need the following two facts:

1. If a complex number z equals its complex conjugate $\overline{z}$ then $z \in \mathbb{R}$
2. $-i = i^{-1}$

To see 1., consider the following argument: Let z be a complex number so that $z = a + bi$. Then $\overline{z} = a - bi$. So $z = \overline{z} \Rightarrow a + bi = a - bi$. Subtracting a from both sides of the right hand side of this implication gives us $bi = -bi$ or $2bi = 0$. Since we know that $2 \neq 0$ and $i \neq 0$ it must be the case that $b = 0$. Said another way: $\mathrm{Im}(z) = 0$ and $\mathrm{Re}(z) = z$. That is, $z \in \mathbb{R}$. So if $z = \overline{z}$ we know that $z \in \mathbb{R}$.

To see 2., notice that $-(i \cdot i) = -i^2 = -(-1) = 1$ and $-i \cdot i = 1 \Rightarrow -i = i^{-1}$.

Now, we want to show that $\overline{i^i} = i^i$, which would show that $i^i \in \mathbb{R}$ by the argument above. So consider:

$$
\begin{aligned}
\overline{i^i} &= \overline{i}^{\overline{i}} && \text{\# definition of complex conjugation [6]} \\
&= (-i)^{-i} && \text{\# since } i = 0 + 1i \text{ and } \overline{0 + 1i} = -1i = -i \\
&= (i^{-1})^{-i} && \text{\# since } -i = i^{-1} \text{ (see 2. above)} \\
&= i^{(-1 \cdot -i)} && \text{\# since } (x^m)^n = x^{mn} \\
&= i^i && \text{\# since } -1 \cdot -i = i
\end{aligned}
$$

So we see that $\overline{i^i} = i^i$ and so $i^i \in \mathbb{R}$. ■

Note that this argument relies on the fact that complex conjugation is an automorphism of $\mathbb{C}$. So one "problem" with this approach is that you would first need to know not only that conjugation is an automorphism of $\mathbb{C}$ but also that the automorphism is equipped with exponentiation. So one could argue that this is proof is incomplete since it may not be true for branches (of the logarithm) other than the principal branch [4].

3 Power Series

We know that the Taylor series [5] for $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$. We also know from Euler's formula [1] that $i = e^{i\frac{\pi}{2}}$ (since $e^{i\frac{\pi}{2}} = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = i$). Note that the same result can be obtained with $x \in \left\{\frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}, \ldots\right\}$, so in general $x = \frac{\pi}{2} \cdot (1 + 4n)$ where $n \in \mathbb{N} \cup \{0\}$.

So we see that

$$\begin{aligned}
i^t &= \left(e^{\frac{i\pi}{2}}\right)^t && \text{\# Euler's formula for } i\text{, raised to the power } t \\
&= e^{\frac{it\pi}{2}} && \text{\# since } (x^m)^n = x^{mn} \\
&= \sum_{k=0}^{\infty} \frac{\left(\frac{it\pi}{2}\right)^k}{k!} && \text{\# since } e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} \text{ and } x = \frac{it\pi}{2}
\end{aligned}$$

If we let $t = i$ we get $i^i = e^{i^2\frac{\pi}{2}} = e^{-\frac{\pi}{2}}$. So $i^i = e^{-\frac{\pi}{2}}$ and since $e^{-\frac{\pi}{2}} \in \mathbb{R}$ we know that $i^i \in \mathbb{R}$. Fortunately this agrees with the result we find in Section 4. ■

4 Euler's Formula

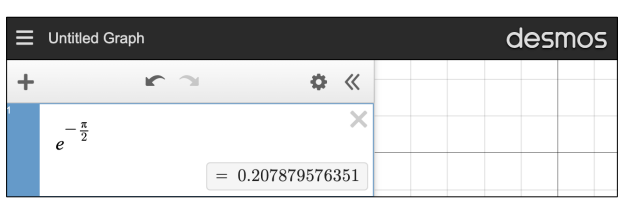
Another approach to answering this question is to consider Euler's formula [1, 3] evaluated at $x = \frac{\pi}{2}$:

$$\begin{aligned}
e^{ix} &= \cos x + i\sin x && \text{\# Euler's formula [1]} \\
&\Rightarrow e^{i\frac{\pi}{2}} = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} && \text{\# set } x = \frac{\pi}{2} \\
&\Rightarrow e^{i\frac{\pi}{2}} = 0 + i \cdot 1 && \text{\# } \cos\frac{\pi}{2} = 0 \text{ and} \sin\frac{\pi}{2} = 1 \\
&\Rightarrow e^{i\frac{\pi}{2}} = i && \text{\# simplify} \\
&\Rightarrow \left(e^{i\frac{\pi}{2}}\right)^i = i^i && \text{\# raise both sides to } i \\
&\Rightarrow e^{\frac{i^2\pi}{2}} = i^i && \text{\# } (x^m)^n = x^{mn} \\
&\Rightarrow e^{-\frac{\pi}{2}} = i^i && \text{\# } i^2 = -1 \\
&\Rightarrow e^{-\frac{\pi}{2}} \in \mathbb{R} \Rightarrow i^i \in \mathbb{R} && \text{\# } i^i \text{ is a real number}
\end{aligned}$$

It is important to notice that this expression is multivalued, and when we evaluate it at $\frac{\pi}{2}$ we are on the principle branch. However, the same result can be obtained with $x \in \left\{\frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}, \ldots\right\}$. In general $x = \frac{\pi}{2} \cdot (1 + 4n)$ for $n \in \mathbb{N} \cup \{0\}$. ■

5 Ok, $i^i \in \mathbb{R}$, but what does it equal

We saw above that $i^i = e^{-\frac{\pi}{2}}$. So $i^i = e^{-\frac{\pi}{2}} \approx 0.20788$.



6 Conclusions

What is missing:

- It would be helpful to have a proof that complex conjugation distributes over complex exponentiation. That is, a proof that $\overline{i^i} = \overline{i}^{\overline{i}}$.

  One argument is that since $\overline{i^i} \in \mathbb{R}$ (as shown above) and the complex conjugate of a real number, it must be that $\overline{i^i} = i^i$. But AFAICU this doesn't show that $\overline{i^i} = \overline{i}^{\overline{i}}$.

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